

Science Connections: Evidence Control

GROW • Science • 40 minutes

I can connect a scientific question to the right idea and use evidence to explain my answer.

Prepare before pupils arrive

Setup: Prepare the three topic headers, six evidence cards and the fixed rover data set.

Safety: No live practical is required. Protect pupils from speed pressure and allow private rehearsal before audience response.

Equipment: Topic cards; evidence cards; three-column response sheet; pencil; optional local card sort.

Safety: This consolidation uses cards and data only. Keep any physical models still unless staff set them up.

Fallback: Use printed cards, speak, point or direct an adult. An exact scribed answer counts equally.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Consolidate forces, space and climate; answer scientific questions.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival • 4 min	State what friction does to movement. Explain one cause of day and night.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter • 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do • 5 min	evidence — what we observed connection — how ideas link cause — what makes a change happen Worked practice question: Why did the rover travel less far on fabric? Name the topic first: forces and friction.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Space answer: day and night happen because Earth rotates, moving a place into and out of sunlight. Climate answer: extra greenhouse gases absorb and re-emit more infrared energy, so the global average rises.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Place each evidence card under Forces, Space or Climate. Say the scientific link, not just the topic name.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Answer the embedded questions: Forces — why did the rover travel less far on fabric? Space — why do day and night happen?	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Use one result to answer a forces, space or climate question. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: The rover travelled less on fabric: 18–20 cm compared with 44–46 cm, because greater friction opposed its motion.

Yes. It answers the question, uses the repeated results and connects them to friction.

Misconception repair

Repaired. A fact may be true but irrelevant; scientific evidence must connect to the question being answered.

Guided checkpoint

Yes. It answers the question, uses the repeated results and connects them to friction. Choose the answer that uses the relevant measurements and explains them with forces.

46, 44, 45 versus 18, 20, 19 cm

FORCES. This evidence supports a forces answer because surface friction changed the travel distance. Which topic explains a difference in an object's movement?

Same load: effort farther from the pivot needs less force

FORCES. This belongs with forces and mechanisms: for the same load and setup, applying effort farther from the pivot reduces the force needed. Look for the topic that includes pushes, friction and mechanisms.

The observer turns into and out of light

SPACE. This supports a space answer: Earth's rotation causes day and night and the apparent Sun movement. Which topic includes Earth, Sun, Moon and rotation?

A sphere stays round from several views

SPACE. This supports the approximately spherical model of the Sun, Earth and Moon. This evidence compares models of celestial bodies.

More outgoing infrared is re-emitted

CLIMATE. This supports a climate answer about the enhanced greenhouse effect. Which topic follows the route of infrared energy through the atmosphere?

300 minutes before; 90 minutes after

CLIMATE. This can support a limited climate-action answer: wasted-light time fell in the local sample. The data check an action intended to reduce unnecessary energy demand.

Model limitation

One short answer cannot show every force, space or climate detail; select only what directly answers the question.

Independent evidence

At least one complete answer with relevant evidence, scientific reasoning and one honest limit.

Supported: Choose one embedded question: rover and friction, day and night, or extra greenhouse gas and warming. Match one claim, evidence card and reason. Use icon-backed choices and the stem: I think ____ because the evidence shows ____.

Standard: Answer two of the three embedded questions from different topics and underline the evidence used. Use Question, Evidence and Reason headings, then check relevance.

Stretch: Answer all three embedded topic questions, compare evidence quality and name one model or data limitation. Explain why an alternative piece of evidence is less relevant or less secure.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_G_W12A_Science_Connections_Explore-1.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.